Milestone 2 Study Plan — Days 13–28

Milestone 2 is on 23 Nov 2026 · Scope: ML, Deep Learning, NLP, Transformers, Generative AI

How to use this plan

For every day, study the topics the evening before the class (a 30-minute preview), then do the code and self-check after class. Each day below lists **Study** (what to understand), **Code** (what to write yourself, with no copy-paste) and **Check yourself** (questions to answer from memory).

- Concept first (30 min): understand the idea in plain words.
- Code (40 min): write it from a blank file.
- Recall (20 min): close everything and answer the check questions aloud or on paper.
- Keep one page of notes per day. Those pages become your revision sheet.

Dates	Days	Focus
31 Oct – 6 Nov	13–18	Machine Learning and start of Deep Learning
9 – 15 Nov	19–23	CNN advanced, RNN/LSTM, GANs
16 – 20 Nov	24–28	NLP, Transformers, Generative AI
21 – 22 Nov	–	Full revision and timed mock
23 Nov	29	Milestone Assessment 2

Day 13 · Fri 30 Oct — ML foundations and pipeline

Study

- Supervised vs unsupervised learning, features and labels, train/validation/test split.
- Pipeline steps: load, clean, encode, scale, split, train, evaluate, save.
- Classification metrics: accuracy, precision, recall, F1, confusion matrix, ROC-AUC. Regression metrics: MAE, MSE, RMSE, R-squared.
- Cross-validation, and why accuracy alone misleads on imbalanced data.
- Saving a model (joblib/pickle) and serving it as an API (you already know FastAPI, so a /predict endpoint is the same idea).

Code

- Build a full scikit-learn Pipeline with ColumnTransformer, train on a CSV, print a classification report, save with joblib, reload and predict.

Check yourself

- When is recall more important than precision?
- Why must the scaler be fitted on training data only?

Day 14 · Mon 2 Nov — Decision Tree, Random Forest, SVM

Study

- Decision Tree: Gini vs entropy, how a split is chosen, depth and overfitting, max_depth and min_samples_leaf, feature_importances_.
- Random Forest: bagging, bootstrap samples, random feature subsets, n_estimators, out-of-bag score.
- SVM: margin, support vectors, C, kernels (linear, RBF, polynomial), gamma, and why scaling is required.

Code

- Train all three on one dataset, compare accuracy, and plot the tree with plot_tree.

Check yourself

- Why is a Random Forest less prone to overfitting than one tree?
- What does a high C do in SVM?

Day 15 · Tue 3 Nov — Model comparison, tuning, bias-variance

Study

- When to use which classical model (linear, tree, forest, SVM, KNN).
- Bias vs variance, underfitting vs overfitting, learning curves, regularization (L1 lasso, L2 ridge).
- GridSearchCV vs RandomizedSearchCV.

Code

- Loop over several models with cross_val_score, then tune the best one with GridSearchCV.

Check yourself

- Training accuracy 99%, test accuracy 70%: what is wrong and what are three fixes?

Day 16 · Wed 4 Nov — Feature engineering and ensembles

Study

- Encoding (one-hot, ordinal), scaling (StandardScaler vs MinMaxScaler), missing values, feature selection.
- PCA: purpose, explained variance ratio, choosing components.
- Bagging vs boosting (AdaBoost, Gradient Boosting, XGBoost), voting and stacking.
- Improvement strategies: tuning, class weights, handling imbalance, error analysis.

Code

- Apply PCA to a dataset and plot explained variance. Compare RandomForest vs GradientBoosting.

Check yourself

- Bagging reduces what? Boosting reduces what?

Day 17 · Thu 5 Nov — Neural networks and ANN with Keras

Study

- Neuron, weights, bias, activation functions (ReLU, sigmoid, tanh, softmax) and where each is used.
- Forward pass, loss, backpropagation, gradient descent, Adam, learning rate, batch size, epochs.
- Dropout, batch normalization, early stopping. How to count parameters in a Dense layer.

Code

- MNIST in Keras: load, divide by 255, Flatten, Dense(128, relu), Dense(10, softmax), compile with sparse_categorical_crossentropy, fit, evaluate, predict one image.

Check yourself

- Why softmax in the output layer, and which loss goes with it?
- Parameters in Dense(128) after 784 inputs?

Day 18 · Fri 6 Nov — CNN theory and Weekly Assessment 2

Study

- Convolution, filters, stride, padding, pooling, feature maps. Output size = (n − f + 2p) / s + 1.

- LeNet (digits, early), AlexNet (ReLU, dropout, GPUs), VGG (stacked 3×3 filters, very deep).
- Why CNNs beat plain ANNs on images (parameter sharing, local patterns).

Code

- CIFAR-10 classifier: Conv2D, MaxPooling2D, Flatten, Dense, plus data augmentation.

Check yourself

- Compute the output shape for a 28×28 input, 3×3 filter, stride 1, no padding.

Day 19 · Mon 9 Nov — Advanced CNNs, transfer learning, object detection

Study

- ResNet (skip connections solve vanishing gradients), Inception, MobileNet (depthwise separable convolution), EfficientNet.
- Transfer learning: feature extraction (freeze base) vs fine-tuning (unfreeze top layers with a small learning rate), include_top=False.
- Object detection: YOLO (single pass, fast), R-CNN family (region proposals, more accurate), SSD. Terms: bounding box, IoU, non-max suppression, mAP.

Code

- Fine-tune MobileNetV2 or ResNet50 on a small image dataset.

Check yourself

- YOLO vs Faster R-CNN: the trade-off in one line.

Day 20 · Tue 10 Nov — Segmentation, visualization, CV project

Study

- Semantic vs instance segmentation. U-Net (encoder–decoder with skip connections). Mask R-CNN (adds a mask branch to Faster R-CNN).
- CNN visualization: filters, feature maps, Grad-CAM, saliency maps.
- Steps of an end-to-end computer vision project.

Code

- Plot feature maps of a trained CNN layer. Write a short end-to-end notebook outline for a CV project.

Check yourself

- Why does U-Net need skip connections?

Day 21 · Wed 11 Nov — RNN, vanishing gradient, LSTM

Study

- RNN: hidden state, processing sequences, backpropagation through time.
- Vanishing and exploding gradients, gradient clipping.
- LSTM: forget, input and output gates and the cell state. GRU as a lighter version. Bidirectional RNN/LSTM.

Code

- Keras LSTM and Bidirectional(LSTM) for sentiment on IMDB, with Embedding layer.

Check yourself

- What does the forget gate decide?
- Why does an LSTM remember longer than a plain RNN?

Day 22 · Thu 12 Nov — Seq2seq, attention, time-series with LSTM

Study

- Encoder–decoder architecture and the context-vector bottleneck.
- Attention: scores, softmax weights, weighted context (Bahdanau/Luong idea).
- Time-series forecasting: windowing, scaling, no shuffling, inverse-transforming predictions, RMSE.

Code

- Forecast a series (e.g. stock or temperature): create windows, train an LSTM, plot predictions vs actual.

Check yourself

- Why does attention help long sentences?

Day 23 · Fri 13 Nov — GANs, VAEs, Weekly Assessment 3

Study

- GAN: generator vs discriminator, adversarial training, mode collapse, instability. Variants: DCGAN, conditional GAN, CycleGAN, StyleGAN, WGAN.
- VAE: encoder outputs mean and log-variance, sampling in latent space, reparameterization trick, loss = reconstruction + KL divergence.

- Compare GAN vs VAE vs diffusion (quality, stability, diversity, speed).

Code

- Skim a small DCGAN or VAE on MNIST and be able to point to the generator, discriminator and loss.

Check yourself

- What is mode collapse? Why is a VAE output usually blurrier than a GAN output?

Day 24 · Mon 16 Nov — NLP basics and classical pipelines

Study

- Tokenization, stopwords, stemming vs lemmatization, Bag of Words, TF-IDF.
- Word embeddings: Word2Vec (CBOW vs skip-gram), GloVe, FastText.
- Classical pipeline: TF-IDF + Logistic Regression or Naive Bayes. Why sequence models led to Transformers.

Code

- Text classifier with TfidfVectorizer + LogisticRegression, and a Word2Vec similarity query.

Check yourself

- What does TF-IDF down-weight and why?

Day 25 · Tue 17 Nov — Attention, Transformers, BERT

Study

- Self-attention: Query, Key, Value and attention = softmax($Q \cdot K^T / \sqrt{d_k}$) · V. Multi-head attention, positional encoding.
- Transformer block: attention, feed-forward, residual connections, layer norm. Encoder vs decoder.
- Why Transformers work: parallel training and long-range dependencies.
- BERT: bidirectional encoder, masked language modelling, [CLS] token, fine-tuning. Variants: RoBERTa, DistilBERT, ALBERT. Optimization: distillation, quantization, pruning.

Code

- Hugging Face: load a tokenizer and model, run a pipeline, and fine-tune or run BERT for classification.

Check yourself

- Why divide by √d_k? Why is positional encoding needed?

Day 26 · Wed 18 Nov — LLM applications and APIs

Study

- Summarization (extractive vs abstractive), sentiment analysis, conversational AI, multilingual models.
- LLM landscape: closed vs open models, context window, tokens.
- API integration: messages, system prompt, temperature, max tokens, streaming, error handling, cost.

Code

- Call an LLM API for summarization and sentiment, with try/except and different temperatures.

Check yourself

- What does temperature change in the output?

Day 27 · Thu 19 Nov — Generative AI intro and diffusion (Mock Assessment 1)

Study

- Discriminative vs generative models, types of generative models.
- Diffusion: forward process adds noise, reverse process denoises, timesteps, U-Net denoiser, latent diffusion.
- Stable Diffusion parts (VAE, U-Net, text encoder) and classifier-free guidance. DALL-E at the idea level.

Code

- Run a diffusers pipeline once, or write the steps from memory.

Check yourself

- Why is diffusion more stable to train than a GAN?

Day 28 · Fri 20 Nov — Image and video generation (Mock Assessment 2)

Study

- Prompt, negative prompt, steps, guidance scale, seed, image-to-image.

- Video generation: temporal consistency, the main challenges.

Code

- Generate three images changing only the guidance scale and steps, and note the effect.

Check yourself

- What happens when guidance scale is too high?

Final revision — 21–22 Nov

- Saturday: reread your daily notes and rebuild your one-page cheat sheet per module (ML, DL, NLP, GenAI).
- Write from memory: sklearn pipeline, Keras ANN and CNN, LSTM, transfer learning.
- Sunday: take a timed mock using the mock assessment questions, then fix the weakest two topics.
- Sleep early on 22 Nov. Last-minute cramming helps less than rest.

Assessment-day tips

- Read each question fully. Start with the one you know best.
- Write clean, runnable code with imports, then run it before submitting.
- Name variables clearly and add a short comment per step.

Part 2 — Day-by-day calendar (31 Oct – 23 Nov)

Weekday = about 2 hours after training. Weekend = about 4–5 hours in blocks with breaks. Tick off each block. If you miss a day, move its Code block to the next weekend instead of skipping it.

Weekday pattern: (1) 20 min revise what was taught today, (2) 50 min code from a blank file, (3) 20 min recall with notes closed, (4) 20 min preview of tomorrow's topics.

Sat 31 Oct — Day 13 deep dive (ML pipeline)

- 60 min: metrics. Write the formulas for precision, recall, F1 and RMSE in your own words, then explain a confusion matrix with a made-up example.

- 90 min: build a full scikit-learn Pipeline on any CSV (for example Titanic): impute, one-hot encode, scale, split, train, classification report.
- 45 min: cross-validation, then save the pipeline with joblib, reload it and predict on one new row.
- 30 min: wrap the saved model in a small FastAPI /predict endpoint.
- 20 min: recall. Explain data leakage and why the scaler is fitted on training data only.

Sun 1 Nov — Day 14 pre-learning (Tree, Forest, SVM)

- 60 min: Decision Tree. Learn Gini and entropy, then train one, change max_depth from 2 to 20 and watch train vs test accuracy.
- 45 min: Random Forest. Learn bagging and feature randomness, check n_estimators and feature_importances_.
- 45 min: SVM. Learn margin, C, gamma and kernels, and see what happens without scaling.
- 30 min: compare all three on one dataset in a single results table.
- 20 min: recall the three check questions from Day 14.

Mon 2 Nov — Day 14 class day

- 20 min: revise class notes and fill gaps from yesterday's learning.
- 50 min: redo the three-model comparison from a blank file, then plot the tree with plot_tree.
- 20 min: recall. Why does a forest overfit less than one tree?
- 20 min: preview Day 15 (bias–variance, GridSearchCV).

Tue 3 Nov — Day 15 (model comparison and tuning)

- 20 min: revise bias–variance, underfitting vs overfitting, L1 vs L2.
- 50 min: loop over 4 models with cross_val_score, tune the best one with GridSearchCV, then draw a learning curve.
- 20 min: recall. Training 99%, test 70%: name three fixes.
- 20 min: preview Day 16 (PCA, encoding, boosting).

Wed 4 Nov — Day 16 (feature engineering and ensembles)

- 20 min: revise scaling, encoding, missing-value strategies.

- 50 min: apply PCA and plot explained variance. Compare RandomForest, GradientBoosting and a VotingClassifier.
- 20 min: recall bagging vs boosting in one sentence each.
- 20 min: preview Day 17 (neurons, activations, Keras basics).

Thu 5 Nov — Day 17 (ANN with Keras)

- 20 min: revise activations, loss functions, backpropagation idea, optimizers.
- 50 min: write the MNIST ANN from a blank file. Add Dropout and EarlyStopping and compare results.
- 20 min: recall. Count the parameters of Dense(128) after 784 inputs, and explain why softmax pairs with its loss.
- 20 min: preview Day 18 (convolution, pooling, output-size formula).

Fri 6 Nov — Day 18 (CNN theory, Weekly Assessment 2)

- 20 min: revise convolution, stride, padding, pooling. Practise the output-size formula on 5 examples.
- 50 min: build the CIFAR-10 CNN with Conv2D, MaxPooling2D, Flatten and Dense, with augmentation.
- 20 min: recall LeNet vs AlexNet vs VGG in one line each.
- Keep the evening lighter. Your priority tonight is Weekly Assessment 2.

Sat 7 Nov — Consolidation 1 (ML and first Deep Learning)

- 90 min: write from memory the sklearn pipeline, then the three-model comparison with tuning.
- 90 min: write from memory the Keras ANN and the CNN. Note every place you had to look something up.
- 45 min: make a one-page cheat sheet for Days 13–18 (metrics, models, activations, CNN formulas).
- 30 min: ask Copilot for 10 exam-style questions on Days 13–18 and answer them without notes.

Sun 8 Nov — Preview of week 2 and rest

- 90 min: transfer learning. Fine-tune MobileNetV2 on a small image set: first freeze the base, then unfreeze the top layers with a small learning rate.

- 45 min: read the idea of ResNet skip connections, YOLO vs Faster R-CNN, IoU and non-max suppression.
- 30 min: recall using the Day 19 check question.
- Take the rest of the day off. Rest counts as part of the plan.

Mon 9 Nov — Day 19 (advanced CNNs, transfer learning, detection)

- 20 min: revise ResNet, MobileNet, Inception, detection terms.
- 50 min: redo the transfer learning notebook from a blank file and compare feature extraction vs fine-tuning accuracy.
- 20 min: recall. Explain IoU and non-max suppression in your own words.
- 20 min: preview Day 20 (U-Net, Grad-CAM).

Tue 10 Nov — Day 20 (segmentation and visualization)

- 20 min: revise semantic vs instance segmentation, U-Net, Mask R-CNN.
- 50 min: plot feature maps from a CNN layer and write a Grad-CAM outline. Draw the U-Net shape on paper.
- 20 min: recall. Why does U-Net need skip connections?
- 20 min: preview Day 21 (RNN and the vanishing gradient).

Wed 11 Nov — Day 21 (RNN, LSTM)

- 20 min: revise hidden state, vanishing gradient, the three LSTM gates.
- 50 min: IMDB sentiment with Embedding + LSTM, then replace with Bidirectional(LSTM) and compare.
- 20 min: recall. What does each LSTM gate decide?
- 20 min: preview Day 22 (seq2seq, attention, time series).

Thu 12 Nov — Day 22 (seq2seq, attention, forecasting)

- 20 min: revise the encoder–decoder bottleneck and how attention fixes it.
- 50 min: time-series forecasting. Window the data, scale, train an LSTM, inverse-transform predictions and plot actual vs predicted.
- 20 min: recall. Why must time series not be shuffled?
- 20 min: preview Day 23 (GANs, VAEs).

Fri 13 Nov — Day 23 (GANs, VAEs, Weekly Assessment 3)

- 30 min: revise generator vs discriminator, mode collapse, VAE loss and the reparameterization trick.
- 30 min: write a comparison table of GAN vs VAE vs diffusion from memory.
- Light evening. Your priority tonight is Weekly Assessment 3.

Sat 14 Nov — Consolidation 2 (Deep Learning)

- 90 min: rebuild from memory a CNN with transfer learning and an LSTM forecaster.
- 60 min: rewrite your cheat sheet for Days 19–23 (architectures, detection terms, gates, GAN/VAE).
- 60 min: 10 Copilot questions on Days 17–23, answered without notes. Mark weak topics.
- 45 min: study NLP preview: tokenization, stemming vs lemmatization, TF-IDF.

Sun 15 Nov — NLP and Transformer preview

- 60 min: build TF-IDF + Logistic Regression text classifier and a Word2Vec similarity query.
- 60 min: understand self-attention: Q, K, V and $\text{softmax}(QK^T/\sqrt{d_k})V$. Work one tiny example with 3 words by hand.
- 30 min: recall. Why is positional encoding needed?
- Short evening off.

Mon 16 Nov — Day 24 (NLP basics)

- 20 min: revise Bag of Words, TF-IDF, Word2Vec (CBOW vs skip-gram), GloVe.
- 50 min: redo the TF-IDF classifier from scratch and add a simple embedding-based similarity search.
- 20 min: recall what TF-IDF down-weights and why.
- 20 min: preview Day 25 (Transformer blocks, BERT).

Tue 17 Nov — Day 25 (Transformers, BERT)

- 20 min: revise multi-head attention, residual connections, layer norm, encoder vs decoder.
- 50 min: load a BERT tokenizer and model with Hugging Face. Run a sentiment pipeline, then see the token IDs and the [CLS] output.

- 20 min: recall. Why divide by √d_k? What is masked language modelling?
- 20 min: preview Day 26 (LLM APIs).

Wed 18 Nov — Day 26 (LLM applications and APIs)

- 20 min: revise summarization types, tokens, context window, temperature.
- 50 min: call an LLM API for summarization and sentiment. Add error handling and try temperatures 0, 0.7 and 1.
- 20 min: recall. What does temperature change?
- 20 min: preview Day 27 (diffusion).

Thu 19 Nov — Day 27 (GenAI intro, diffusion, Mock Assessment 1)

- 20 min: revise forward noising, reverse denoising, latent diffusion, classifier-free guidance.
- 40 min: Mock Assessment 1. Treat it as the real exam: no notes, timed.
- 30 min: review mistakes and write each fix as one line.
- 20 min: preview Day 28.

Fri 20 Nov — Day 28 (image/video generation, Mock Assessment 2)

- 20 min: revise prompt, negative prompt, steps, guidance scale, seed.
- 40 min: Mock Assessment 2 under timed conditions.
- 30 min: list the three weakest topics from both mocks.

Sat 21 Nov — Final revision day

- 90 min: rewrite from memory the sklearn pipeline, Keras ANN, CNN with transfer learning, LSTM and a Hugging Face pipeline.
- 60 min: fix the three weakest topics from the mocks using your Copilot Q&A prompt.
- 60 min: read all four cheat sheets (ML, DL, NLP, GenAI) once, aloud.
- 30 min: pack your setup for exam day (laptop charged, IDE and libraries working, internet check).

Sun 22 Nov — Light day

- 90 min: one timed full mock covering mixed topics from Days 13–28.
- 30 min: review only your cheat sheets.

- Stop by early evening. Sleep early.

Mon 23 Nov — Milestone Assessment 2

- Read each question twice and start with the one you know best.
- Write clean, runnable code with imports and run it before submitting.
- Keep a few minutes at the end to check output shape, variable names and comments.